# Homework 2 · Instructor solutions

Welfare and risk · Lectures 3 and 4

**Order and assessment.** Part I is the ordinary assessed training set, out of 100: Training 1 is 30 points, Training 2 is 35 and Training 3 is 35. It appears before Part II as requested. Part II is the 120-minute formative mock, also out of 100: 50 true/false, 20 consumer welfare, 20 insurance and 10 short answers. Do not combine the two totals. Only Part I contributes to the assessed homework score; the three homeworks remain equally weighted within the course's 20% component. No deadline has been supplied.

**Scope of this preparation draft.** Lecture 3 covers consumer duality and Slutsky. Lecture 4 is being planned to start with CV, EV and CS, then introduce expected utility, risk aversion and insurance, following the course syllabus's risk block. Its slides are not yet in this repository. This draft uses that combined scope. FOSD is used in training and one mock claim; the SOSD coupling is ungraded additional practice only.

**Mock marking.** Each true/false claim is worth 5 points, normally 1 for the verdict and 4 for a correct justification; a bare label earns zero. Accept equivalent valid counterexamples and give partial credit for useful intermediate reasoning. The key is **T, F, F, F, T; T, F, T, F, F**. The 120-minute paper is diagnostic: concise rigorous arguments suffice.

**Suggested workload.** Part I has no time limit. A student who already knows the tools should allow roughly two hours for the required training and two for the mock, with extra time for review. The optional SOSD exercise is not required or marked. This is a planning estimate, not a student time restriction.

# Training 1 · 30 points

Write $P(p) = p_1^{1/4} p_2^{3/4}$. This is a valid expenditure function, for example for the normalised Cobb–Douglas index

$$u(x) = \frac{4}{3^{3/4}} x_1^{1/4} x_2^{3/4}.$$

Students need not reconstruct that primitive index.

**(a), 6 points.**

$$h_1 = \frac{\bar{u}P}{4p_1}, \qquad h_2 = \frac{3\bar{u}P}{4p_2}, \qquad p_1 h_1 + p_2 h_2 = \bar{u}P = e.$$

Suggested split: 2 points per quantity, 2 for the spending identity.

**(b), 8 points.** Solving $m = e(p, \bar{u})$ gives $v = m/P$. Since

$$v_m = \frac{1}{P}, \quad v_{p_1} = -\frac{m}{4p_1 P}, \quad v_{p_2} = -\frac{3m}{4p_2 P},$$

Roy yields $x_1 = m/(4p_1)$ and $x_2 = 3m/(4p_2)$. Substituting $\bar{u} = m/P$ into h gives exactly those quantities. Suggested split: 2 for v, 4 for Roy and the demands, 2 for the parameter match.

**(c), 10 points.** Initially $P^0 = 1$, $x^0 = (4, 12)$ and $u^0 = 16$. At the new prices $P^1 = 2$, $x^1 = (1/4, 12)$ and $u^1 = 8$. With positive-loss conventions,

$$CV = e(p^1, u^0) - 16 = 32 - 16 = 16,$$

$$EV = 16 - e(p^0, u^1) = 16 - 8 = 8,$$

$$L_{CS} = \int_1^{16} \frac{4}{s}\, ds = 4 \log 16 = 16 \log 2 \approx 11.0904.$$

Thus $EV < L_{CS} < CV$. Both goods are normal. At the compensated new income 32, the bundle is $(1/2, 24)$, which restores utility 16. Suggested split: 2 for bundles/utilities, 3 CV, 3 EV, 2 CS and comparison.

**(d), 6 points.** The old basket costs $16 \cdot 4 + 12 = 76$ at new prices. Making it affordable adds $76 - 16 = 60$, whereas CV adds only 16. With income 76 and the new prices, indirect utility is $76/2 = 38 > 16$; reoptimisation selects $(19/16, 57)$. Keeping a basket affordable generally overcompensates compared with the cheapest way to restore utility. Suggested split: 2 for the payment, 2 for the CV comparison, 2 for new utility and its interpretation.

## Training 2 · 35 points

**(a), 6 points.** Both means equal 68. For $4 \le w < 36$, $F_A(w) = 0 < F_B(w) = 1/2$, while for $100 \le w < 132$, $F_A(w) = 1 > F_B(w) = 1/2$. The CDFs cross, so neither lottery FOSD-dominates the other. Award 2 for the means, 4 for the two inequality regions and conclusion.

**(b), 8 points.** Under the square-root index,

$$\mathbb{E}b(A) = 8, \quad CE_A = 64, \quad RP_A = 4,$$

$$\mathbb{E}b(B) = 1 + \sqrt{33}, \quad CE_B = 34 + 2\sqrt{33}, \quad RP_B = 34 - 2\sqrt{33}.$$

Numerically B's CE is about 45.4891 and its risk premium about 22.5109. Award 4 points per lottery, with credit for correct intermediate calculations.

**(c), 7 points.** Under log utility, $\mathbb{E}g(A) = \log 60$, so $CE_A = 60$. The square-root consumer prefers A to sure 62 because $64 > 62$; the log consumer prefers sure 62 because $60 < 62$. Although log and square root rank certain wealth levels the same way, one is not a positive affine transformation of the other. Nonlinear transformation before taking an expectation changes risk attitudes. Award 2 for the log CE, 3 for the opposite rankings, 2 for the representation argument.

**(d), 6 points.**

$$\mathcal{A}_b(w) = \frac{1}{2w}, \quad \mathcal{R}_b(w) = \frac{1}{2}, \qquad \mathcal{A}_g(w) = \frac{1}{w}, \quad \mathcal{R}_g(w) = 1.$$

Log has higher absolute risk aversion at every positive wealth. Award 1 point per coefficient and 2 for the comparison and interpretation.

**(e), 8 points.** For a finite positive-wealth lottery and increasing concave b, $\mathbb{E}b(W) \le b(\mathbb{E}W)$ by Jensen. As $b(CE) = \mathbb{E}b(W)$ and b is increasing, $CE \le \mathbb{E}W$. With strict concavity on an interval containing the support and at least two distinct outcomes of positive probability, Jensen is strict, hence the risk premium is positive. Award 4 for Jensen, 2 for applying monotonicity, 2 for the strictness conditions.

**Ungraded extension.** The marginal probability of $B = 4$ is $(1/2)(3/4)+(1/2)(1/4) = 1/2$. The conditional means are $(3/4)4 + (1/4)132 = 36$ and $(1/4)4 + (3/4)132 = 100$. Thus B is a mean-preserving spread of A under this coupling. For any increasing concave $\phi$,

$$\mathbb{E}[\phi(B) \mid A] \le \phi(\mathbb{E}[B \mid A]) = \phi(A),$$

so $\mathbb{E}\phi(B) \le \mathbb{E}\phi(A)$. This is SOSD without FOSD. It has no points and no submission requirement.

# Training 3 · 35 points

**(a), 6 points.** Final wealth is $w_N = 100 - cI$ with probability $3/4$ and $w_L = 16 + (1-c)I$ with probability $1/4$. Hence

$$U_c(I) = \frac{3}{4}\log(100 - cI) + \frac{1}{4}\log(16 + (1-c)I), \quad 0 \le I \le 84.$$

For either specified price both wealth levels remain positive throughout the interval. The objective is continuous on a compact interval and strictly concave, so a unique optimum exists. Award 3 for wealths and probabilities, 1 for the objective and interval, 2 for existence and uniqueness.

**(b), 7 points.** At $c = 1/4$, expected final wealth is 79 for every coverage. Full coverage $I = 84$ equalises wealth at 79. Strict concavity and Jensen give $\mathbb{E}\log W \le \log 79$, with equality only for certain wealth. Thus full insurance is uniquely optimal. Award 2 for the constant mean, 2 for the bundle, 3 for Jensen and uniqueness.

**(c), 10 points.** At $c = 1/2$,

$$U'(I) = -\frac{3/8}{100 - I/2} + \frac{1/8}{16 + I/2}.$$

The FOC is $100 - I/2 = 3(16 + I/2)$, giving $I^* = 26$ and $(w_N, w_L) = (87, 29)$. It is strictly inside $[0, 84]$; strict concavity makes it the unique global maximiser. Equivalently, $U'(0) > 0$ and $U'(84) < 0$. Award 3 for the derivative/FOC, 3 for I, 2 for wealths, 2 for global verification.

**(d), 12 points.** Replace 100 by W and 16 by $W - 84$. The interior FOC becomes

$$W - I/2 = 3(W - 84 + I/2) \quad \Rightarrow \quad I = 126 - W.$$

For $W > 84$ this candidate is always below 84; it is positive exactly when $W < 126$. Therefore

$$I^*(W) = \begin{cases} 126 - W, & 84 < W < 126, \\ 0, & W \ge 126. \end{cases}$$

At $W = 126$, the zero-coverage bound has zero first derivative but is still the unique optimum by strict concavity. Log utility has decreasing absolute risk aversion $1/w$; here a richer household buys less protection against the same fixed loss at the same loaded price. Fair insurance in (b) remains full, so decreasing coverage is not a universal statement about every premium rule. Award 4 for the general FOC/candidate, 4 for the regimes and boundary, 4 for ARA and the economic explanation.

# Mock 1. True/false solutions · 50 points

## 1. True

The target-feasible set K is fixed as prices change. For $0 \le t \le 1$,

$$\min_{x\in K}(tp+(1-t)q)\cdot x \ge t\min_{x\in K} p\cdot x + (1-t)\min_{x\in K} q\cdot x.$$

No convexity of K is used.

## 2. False

Take $u(x) = \max\{2x_1, x_2\}$. It is continuous and LNS. Its value is

$$v(p,m) = m\max\{2/p_1, 1/p_2\}.$$

At $p=(2,1)$, the income-only value is m and $v_m = 1 > 0$, but $X = \{(m/2,0),(0,m)\}$. Price differentiability fails.

## 3. False

For $u(x) = (1+x_1)x_2$ and $m > p_1$, the unique interior demand is

$$x_1 = \frac{m-p_1}{2p_1}, \qquad x_2 = \frac{m+p_1}{2p_2}.$$

These formulas follow by taking logs after excluding $x_2 = 0$ and solving the strictly concave log problem. They are smooth in this open regime, but $x_{1,p_2} = 0$ while $x_{2,p_1} = 1/(2p_2) > 0$.

## 4. False

Homogeneity gives $Sp = 0$ and $p \neq 0$, hence $p^\top Sp = 0$. Strict negative definiteness is impossible. A fully specified example is Seminar 3's $u = \sqrt{x_1} + 2\sqrt{x_2}$ at $p=(1,4)$, $m=8$, where $S = \begin{pmatrix} -4 & 1 \\ 1 & -1/4 \end{pmatrix}$.

## 5. True

For any minimiser h and any positive price vector q,

$$e(q,\bar{u}) \le q\cdot h = e(p,\bar{u}) + (q-p)\cdot h.$$

If e is differentiable at p, perturb q in either sign of every direction: the inequality forces $h = \nabla_p e(p,\bar{u})$. Every minimiser equals the same vector. Existence follows from continuity, positive prices and target attainability.

## 6. True

Expected transformed utility is $3\mathbb{E}b(W)+7$, so all comparisons are preserved. The same certain wealth solves the transformed CE equation. Also $-\widetilde{b}''/\widetilde{b}' = -3b''/(3b') = -b''/b'$.

## 7. False

Take $b(w)=\sqrt{w}$. Compare equally likely wealths 1 and 9 with sure wealth $9/2$. Expected b is 2 for the lottery and $3/\sqrt{2} > 2$ for the sure wealth. After squaring b, expected utility is expected wealth: 5 for the lottery, $9/2$ for the sure amount. The preference reverses.

## 8. True

For a common finite support $w_1 < \cdots < w_n$,

$$\mathbb{E}b(W) = b(w_1) + \sum_{k=1}^{n-1}[b(w_{k+1}) - b(w_k)]\,\mathsf{Pr}(W > w_k).$$

FOSD means all the relevant tail probabilities of A are at least those of B. The increments of increasing b are nonnegative, so the expected-utility inequality follows. An equivalent monotone-coupling or definition-based argument is acceptable; concavity is not used.

## 9. False

Log utility has DARA, $\mathcal{A} = 1/w$. Let initial wealth be 9 and a loss of 8 occur with probability $1/2$. At a fair unit premium $1/2$, any coverage has mean wealth 5, while full coverage makes wealth exactly 5. Strict concavity of log and Jensen make full coverage uniquely optimal.

## 10. False

Use the increasing concave index

$$b(w) = \begin{cases} w, & 0 < w \le 3, \\ 3 + \mathsf{log}(w-2), & w > 3. \end{cases}$$

It is continuous with matching derivative 1 at 3; its derivative thereafter decreases. Equally likely wealths 1 and 3 have mean and CE both equal to 2, so their risk premium is zero. Yet the index is strictly concave above 3: it strictly prefers sure 5 to equally likely 4 and 6. Local affine behaviour on one lottery's support does not imply global risk neutrality.

# Mock 2 · 20 points

Let $R = \sqrt{p_1} + \sqrt{p_2}$.

**(a), 5 points.**

$$v_m = \frac{1}{R^2}, \qquad v_{p_i} = -\frac{m}{\sqrt{p_i} R^3}, \qquad x_i = -\frac{v_{p_i}}{v_m} = \frac{m}{\sqrt{p_i} R}.$$

The budget identity is $\sum_i p_i x_i = m(\sqrt{p_1} + \sqrt{p_2})/R = m$. Suggested split: 2 for derivatives, 2 for demands, 1 for spending.

**(b), 5 points.** Invert $v(p, e) = \bar{u}$:

$$e(p, \bar{u}) = \bar{u} R^2, \qquad h_i = e_{p_i} = \frac{\bar{u} R}{\sqrt{p_i}}.$$

Suggested split: 2 for expenditure, 3 for both Hicksian quantities.

**(c), 6 points.** At old prices, $R^0 = 2$, $x^0 = (4, 4)$ and $u^0 = 2$. At new prices, $R^1 = 3$, $x^1 = (4/3, 8/3)$ and $u^1 = 8/9$. Therefore

$$CV = e((4, 1), 2) - 8 = 18 - 8 = 10,$$

$$EV = 8 - e((1, 1), 8/9) = 8 - 32/9 = 40/9.$$

Suggested split: 2 for utilities, 2 for each correctly signed monetary measure.

**(d), 4 points.**

$$L_{CS} = \int_1^4 \frac{8}{\sqrt{s}(\sqrt{s} + 1)}\, ds = 16 \int_1^2 \frac{dt}{t + 1} = 16 \log(3/2) \approx 6.4874.$$

Thus $40/9 < L_{CS} < 10$, or $EV < L_{CS} < CV$. The three amounts differ because income effects matter. Suggested split: 3 for the integral and value, 1 for comparison.

# Mock 3 · 20 points

**(a), 4 points.** Uninsured wealth is 144 or 16, equally likely. The mean is 80, expected square-root utility is 8, CE is 64 and the risk premium is 16. Award 1 each.

**(b), 4 points.** A full-insurance lump premium F leaves certain wealth $144 - F$. Indifference requires $\sqrt{144 - F} = 8$, so $F_{\max} = 80$. The expected loss is 64; the extra willingness to pay is 16, the initial risk premium. Award 2 for the maximal fee and 2 for the comparison.

**(c), 4 points.**

$$w_N = 144 - \frac{2I}{3}, \qquad w_L = 16 + \frac{I}{3},$$

$$U(I) = \frac{1}{2}\sqrt{144 - 2I/3} + \frac{1}{2}\sqrt{16 + I/3}, \quad 0 \le I \le 128.$$

Both wealths are positive on this interval. Each square root is strictly concave in the nonconstant affine coverage argument, so $U'' < 0$. Award 2 for state wealths, 1 for the objective, 1 for strict concavity.

**(d), 6 points.**

$$U'(I) = -\frac{1}{6\sqrt{w_N}} + \frac{1}{12\sqrt{w_L}}.$$

The FOC gives $\sqrt{w_N} = 2\sqrt{w_L}$ or $w_N = 4w_L$. Thus $144 - 2I/3 = 4(16 + I/3)$ and $I^* = 40$. The point is inside the allowed interval and strict concavity makes it globally optimal. The state wealths are $352/3$ and $88/3$. Award 3 for solving the FOC, 1 for feasibility/global verification, 2 for the two wealths.

**(e), 2 points.** Since $w_N = 4w_L$, expected square-root utility is $(3/2)\sqrt{88/3}$, whose certainty equivalent is 66, above the uninsured 64. Mean wealth is $220/3 < 80$. The consumer trades some mean wealth for less risk and higher expected utility, without any contradiction. Award 1 for CE, 1 for the interpretation; a separate full mean calculation is not required.

# Mock 4 · 10 points

Model answers below each have three sentences. Accept other valid answers that respect the stated assumptions.

**1. The inflation-proof stipend, 5 points.** The payment keeps the old basket affordable, so the maximum attainable utility cannot fall. Reoptimisation at the new relative prices can raise utility strictly above the old level. Hicksian compensation instead uses $e(p^1, u^0)$, which is no greater than the new cost of the old basket.

Award 2 for affordability and the weak welfare comparison, 2 for the possible strict gain, 1 for distinguishing the minimum expenditure benchmark.

**2. The warranty and the raffle, 5 points.** A strictly increasing globally concave wealth index rules out the independent ticket: its expected net payoff is $-1/2$, so conditional Jensen makes it strictly worse than not buying it. Loaded insurance can still be optimal, so the contradiction comes from the ticket. Explaining both purchases requires changing, for example, monetary-only payoffs, global concavity, or the independence assumption.

Award 2 for the expected payoff and Jensen logic, 2 for identifying the ticket rather than condemning insurance, 1 for a specific assumption change. Simply calling both decisions irrational earns no credit. If T is the ticket's net payoff, the formal argument is

$$\mathbb{E}[b(W+T) \mid W] \le b(W + \mathbb{E}T) = b(W - 1/2) < b(W),$$

using independence, positive final wealth and increasing concavity.